



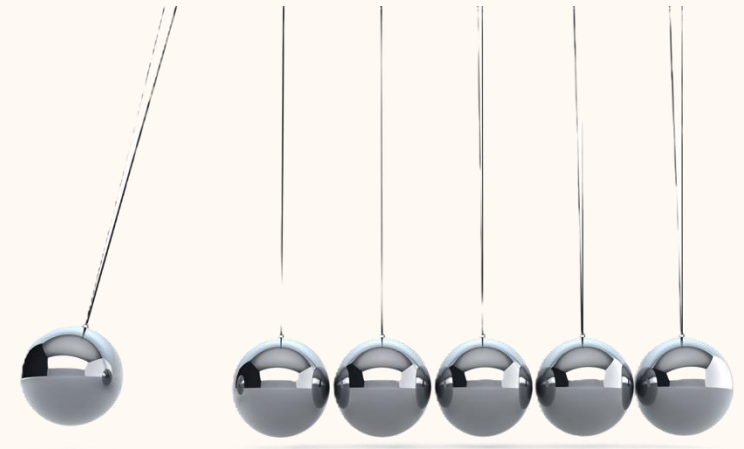
100 Challenge Problems in Pre-University Physics

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Session outline



1. About the book
2. Support for students
3. Problem solving!



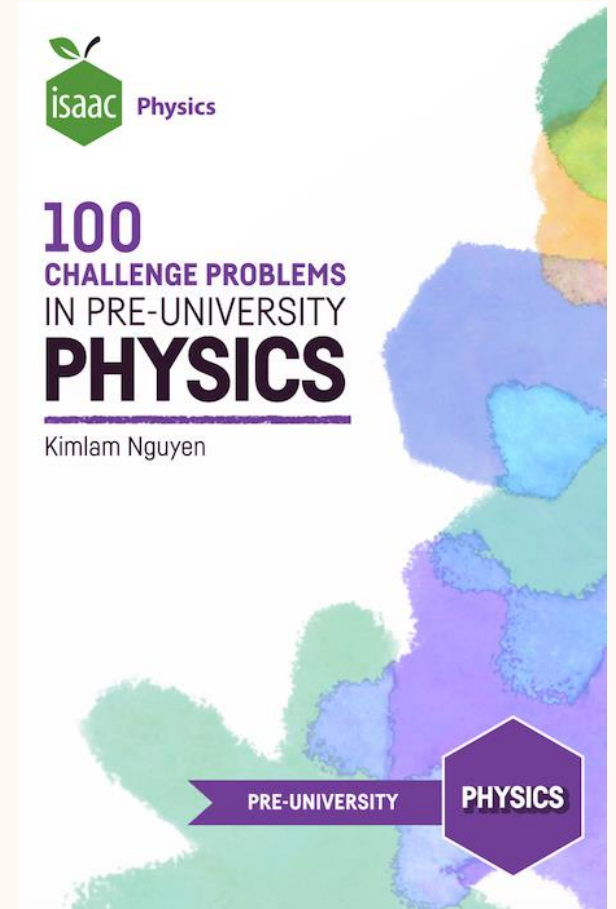
New Isaac Physics book



100 Challenge Problems in Pre-University Physics

is a new collection of problems written for Isaac Science, with some appearing in the British Physics Olympiad.

Release late 2026



Book summary



Features unstructured and extended physics problems, while still using the ideas, topics and techniques taught to Pre-University students.



Extension material for Year 12/13 students, studying both **A Level Physics** and **Maths** (or equivalent) who want to take their physics further.



Further Maths techniques are not required.

Book summary



Solving such problems will help students:

- Apply ideas taught at A Level to unfamiliar problems with a range of difficulties.
- Build mathematical fluency using Pre-University **Maths** skills and applying them to challenging **Physics** problems.
- Develop multiple-step thinking in **Physics**, an essential skill for those progressing to olympiad or university level.

Book structure



100 questions organised:

- by field and topic,
 1. **Mechanics:** Statics, Kinematics, Dynamics, Materials, Circular Motion and SHM
 2. **Electricity:** Resistors, Internal Resistance, Components and Capacitors
 3. **Waves & Particles:** Optics, Wave Motion, Superposition, Quantum and Nuclear
 4. **Thermal:** Heat Capacity and Gases
 5. **Fields:** Gravitational and Electric

Book structure



100 questions organised:

- by difficulty,
 - **C1**: early in the course; uses knowledge from one topic only.
 - **C2**
 - **C3**: multiple topics and/or more advanced mathematical techniques.

Hint structure



Each question has a corresponding hint box with:

1. Concepts

Corresponding online concept page for each topic required

2. Setup

Explanation of first few steps and key equations

3. Discussion

Comments about the problem

Hints for problem 5

Concepts

- Newton's first law
- Friction

Setup

It is useful to draw two diagrams: a sideways view of the slope, and view looking downwards along the normal to the slope. Note that the maximum sideways force is $A mg$. In limiting equilibrium, the frictional force balances the vector sum of $A mg$ sideways and the component of the weight of the phone acting down the slope $mg \sin \theta$. At the point of slipping, the frictional force does not act along the line of greatest slope. Its direction defines the line along which the phone will begin to slip.

Discussion

Check the limits: notice that the minimum value of A is zero when $\mu = \tan \theta$ (recall that this is the condition for limiting equilibrium for a block on a slope), i.e. any sideways disturbance will make it slip. If A is large then the direction of motion is almost horizontal.

Printed book and online version



- Check answers on Isaac
- Hints are provided on Isaac in the usual style:
 1. Concepts
 2. Goal and information given
 3. Diagram
 4. Equations
 5. Hint video

Support for students



Monthly Physics **Mentoring Scheme**

isaacscience.org/pages/mentoring

- A free online scheme focussing on some of these and other challenging Physics problems across Isaac.
- A set of ~8 problems assigned each month.
- Monthly hour-long live webinars on Zoom.



Problem solving!



- Have a go at problems **5** & **21** on the printout
- Read through hints
- Feedback and go through questions